TIANAO LI

https://lukeli0425.github.io tianaoli@u.northwestern.edu https://github.com/Lukeli0425/
Department of Computer Science, McCormick School of Engineering, Northwestern University, Evanston, IL 60208

EDUCATION

Northwestern University — Evanston, IL
Ph.D. Candidate in Computer Science — *Sept. 2023 - Present*
GPA: 3.97/4.0
Research advisor: Emma Alexander

Northwestern University — Evanston, IL
M.S. in Computer Science — *Sept. 2023 - Jun. 2026*

Tsinghua University — Beijing, P.R. China
B.Eng. in Electronic Engineering, Magna Cum Laude — *Aug. 2019 - Jun. 2023*
Curricular Certificate in Astronomy
GPA: 3.85/4.0 (top 15%)
Research advisor: Cheng Ma, Lu Fang

HONORS & AWARDS

- Best Paper Honorable Mention, ICCV 2025 Workshop on Responsible Imaging — *Oct. 2025*
- Best Poster Honorable Mention, Midwest Machine Learning Symposium (MMLS), 2025 — *Jun. 2025*
- Outstanding Graduate, Tsinghua University (top 10%) — *Jun. 2023*
- Scholarship of Comprehensive Excellence, Tsinghua University (top 10%) — *Oct. 2022*
- Scholarship of Comprehensive Excellence, Tsinghua University (top 10%) — *Oct. 2021*
- Scholarship of Social Work, Tsinghua University — *Oct. 2020*

RESEARCH INTERESTS

My research interest is in the field of computational imaging and photography, which lies at the intersection of optics, signal processing, computer vision, and machine learning. Specifically, I study how generative priors, scene representations, and physics-based forward modeling can advance imaging — turning ill-posed measurements into faithful reconstructions of the world. My current applications span astronomical imaging, medical imaging, dynamic scene reconstruction, and monocular depth estimation.

PUBLICATIONS

* Equal contribution.

[1] **Tianao Li**, Tjitske Starkenburg, Yu Sun, Emma Alexander. "**ShuffleFlow: Scalable Posterior Inference for Bayesian Inverse Imaging.**" *IEEE International Conference on Computational Photography (ICCP)*, 2026. **Best Poster Honorable mention**, *Midwest Machine Learning Symposium (MMLS)*, 2025.

[2] Marcos A. Ferreira*, **Tianao Li*** John Mamish*, Josiah Hester, Yaman Sangar, Qi Guo, Emma Alexander. "**SpiderCam: Low-Power Snapshot Depth from Differential Defocus.**" *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026 (**Highlight**).

[3] **Tianao Li**, Manxiu Cui, Cheng Ma, Emma Alexander. "**Coordinate-based Speed of Sound Recovery for Aberration-Corrected Photoacoustic Computed Tomography.**" *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. **Best Paper Honorable mention**, *ICCV 2025 Workshop on Responsible Imaging*.

[4] **Tianao Li**, Emma Alexander. "**Galaxy Image Deconvolution for Weak Gravitational Lensing with Unrolled Plug-and-Play ADMM.**" *Monthly Notices of the Royal Astronomical Society: Letters*, 2023.

EXPERIENCE

Advanced Technology Group, Dolby Laboratories
PhD Research Intern — *Jun. 2026 - Sept. 2026*

NSF-Simons AI Institute for the Sky (SkAI Institute)
Graduate Student Researcher — *Jan. 2025 - Present*
Advisor: Emma Alexander, Tjitske Starkenburg

Bio-Inspired Vision Lab, Northwestern University
Graduate Research Assistant — *Sept. 2023 - Present*
Advisor: Emma Alexander

Biophotonics Lab, Tsinghua University
Undergraduate Research Assistant — *Oct. 2022 - Jun. 2023*
Advisor: Cheng Ma

Bio-Inspired Vision Lab, Northwestern University
Research Intern (remote) — *Apr. 2022 - Feb. 2023*
Advisor: Emma Alexander

SIGMA Lab, Tsinghua University
Research Assistant — *Sept. 2021 - Feb. 2022*
Advisor: Lu Fang

INVITED TALKS

Coordinate-based Speed of Sound Recovery for Aberration-Corrected Photoacoustic Computed Tomography
ICCV 2025 Workshop on Responsible Imaging — *Oct. 2025*

Probabilistic Imaging of Galaxies for Weak Gravitational Lensing
Open SkAI 2025 — *Sep. 2025*

Galaxy Image Deconvolution for Weak Gravitational Lensing with Unrolled Plug-and-Play ADMM
ECE Seminar, Department of Electrical & Computer Engineering, Boston University — *Jul. 2024*
Center for Interdisciplinary Exploration and Research in Astrophysics, Northwestern University — *Jan. 2024*
PKU Computational Scientific Imaging Group, Peking University — *Dec. 2023*
Astro Imaging Workshop 2023, Northwestern University — *Jul. 2023*

ACADEMIC SERVICE

- Reviewer, British Machine Vision Conference (BMVC) 2026
- Reviewer, IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2026
- Reviewer, IEEE Open Journal of Signal Processing (OJSP)
- Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025